ECE 332: Engineering Electromagnetics II

Lecture 4

Mark A. Martin

Portland State University

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Quick Recap

- We completed our discussion of how to use **Faraday's law** to determine the **electromotive force (emf)**

$$V_{\text{emf}} = -N\frac{d\Phi}{dt} = -N\frac{d}{dt}\int_S \mathbf{B}\cdot d\mathbf{s} \tag{1}$$

and the current

$$I = \frac{V_{\text{emf}}}{R + R_i} \approx \frac{V_{\text{emf}}}{R} \tag{2}$$

that a magnetic field **B induces** in a conducting loop with N turns, a load R, and internal resistance R_i whenever the magnetic flux Φ through the loop changes. Here, the **magnetic flux** through the loop (in the direction of $d\mathbf{s}$) is

$$\Phi = \int_S \mathbf{B}\cdot d\mathbf{s}, \tag{3}$$

where S is the surface of the loop and $d\mathbf{s}$ points in a direction normal to the surface.

Quick Recap (Continued)

- For a loop with a single turn, there were **3 cases** to consider.
  1. Time-varying flux through a stationary loop:

$$V_{\text{emf}} = V_{\text{emf}}^{\text{tr}} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} \quad (\textbf{transformer emf}) \tag{4}$$

  2. A loop moves in a stationary magnetic field:

$$V_{\text{emf}} = V_{\text{emf}}^{\text{m}} = \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\boldsymbol{\ell} \quad (\textbf{motional emf}) \tag{5}$$

  3. A loop moves in a time-varying flux:

$$V_{\text{emf}} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} + \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\boldsymbol{\ell} \tag{6}$$

- In Lecture 2, we completed our discussion of Case 1. Last lecture, we started by continuing our discussion of Case 2.

Quick Recap (Continued)

- We considered an example of finding the induced emf and current across a resistor in a **moving loop** in a **stationary non-uniform magnetic field**. Sides with components perpendicular to the plane of the velocity **u** and the magnetic field **B** contributed to the emf and those lying in the plane of the velocity and the magnetic field made no contribution.

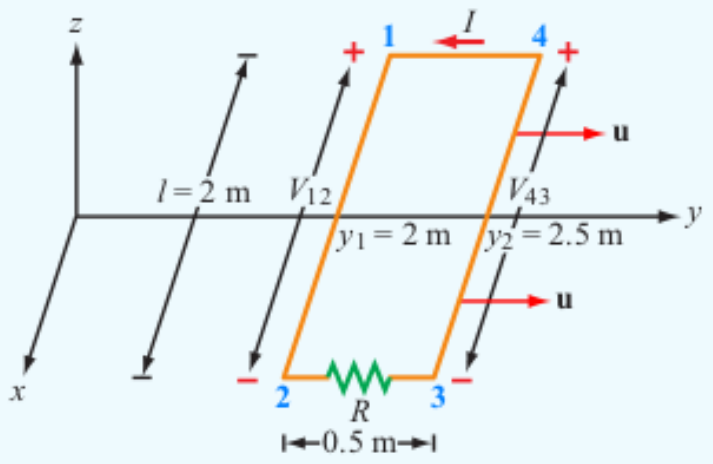


Figure 6-9 Moving loop of Example 6-4.

Quick Recap (Continued)

- An **electromagnetic motor** or an **electromagnetic generator** consists of a conducting loop that rotates in a stationary magnetic field and are essentially the same device.
  - In a motor, the loop contains a voltage source that generate a torque and causes the loop to rotate. It converts electromagnetic power into mechanical power.
  - In a generator, a mechanical torque rotates the loop and a current is generated in the loop. It converts mechanical power into electromagnetic power.

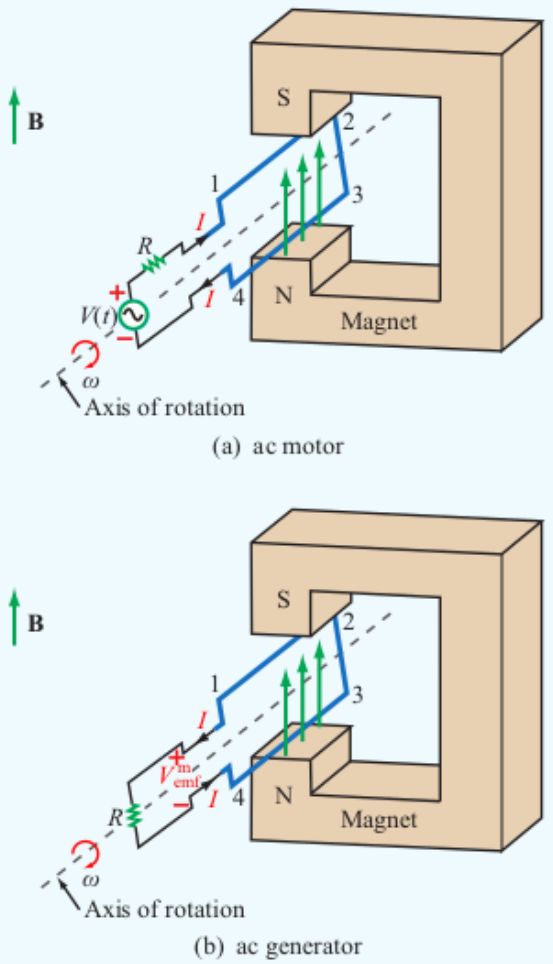


Figure 6-11 Principles of the ac motor and the ac generator. In (a) the magnetic torque on the wires causes the loop to rotate, and in (b) the rotating loop generates an emf.

Quick Recap (Continued)

- **We considered the example of a generator** consisting of a rectangular loop with area A rotating at a constant angular velocity ω, starting from an angle C_0, in a constant magnetic field with magnitude B_0.
  - The **emf generated** in the loop is

$$V_{\text{emf}}^{\text{m}} = A\omega B_0 \sin(\omega t + C_0). \tag{7}$$

  - **We found this result in 2 ways –**
    1. Using the formula for $V_{\text{emf}}^{\text{m}}$ around the loop
    2. Using Faraday's law directly based on the flux Φ through the loop.
  - The formula can be used to **find ways to increase** the generated emf.

Quick Recap (Continued)

- The 3rd **and most general case of applying Faraday's law** is when the conducting loop moves and the magnetic field changes. In this case, the **total emf** can be partitioned into **a transformer and a motional emf** and calculated as

$$V_{\text{emf}} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} + \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}. \tag{8}$$

or **Faraday's law can be used directly** to determine

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt}\int_S \mathbf{B} \cdot d\mathbf{s}, \text{ where } \Phi = \int_S \mathbf{B} \cdot d\mathbf{s}. \tag{9}$$

These are mathematically equivalent. Whichever approach is simpler will be used in a specific context.

Quick Recap (Continued)

- As an example of the most general case, we considered the **same generator with initial angle $C_0 = 0$ and a sinusoidally varying magnetic field**

$$\mathbf{B} = B_0 \cos(\omega t)\hat{\mathbf{z}}. \tag{10}$$

  In this case, a direct application of Faraday's law via Φ showed that

$$V_{\text{emf}} = A\omega B_0 \sin(2\omega t), \tag{11}$$

  which was very similar to the previous example where the magnetic field was constant.

- Faraday's law is one of Maxwell's equations that is different when charges move and magnetic fields change. The other equation that is different in the dynamic case is **Ampère's law**

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}. \tag{12}$$

Quick Recap (Continued)

- For an open surface S with boundary contour C, the **integral form of Ampère's law** can be written as

$$\oint_C \mathbf{H} \cdot d\ell = I = I_c + I_d, \tag{13}$$

where I is the **total current** flowing through the surface,

$$I_c = \int_S \mathbf{J} \cdot d\mathbf{s} \tag{14}$$

is the **conduction current** flowing through the surface, and the **displacement current** is

$$I_d = \int_S \mathbf{J}_d \cdot d\mathbf{s}, \text{ where } \mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}. \tag{15}$$

Here, $\mathbf{J}_d$ is the **displacement current density**.

Quick Recap (Continued)

- The example of an **ideal parallel-plate capacitor** showed that, even though the displacement current does not transport charges, it makes the total current in a circuit continuous.
- An example of a **more realistic wire** showed that I_d can generally be ignored in good conductors.

Boundary Conditions for Electromagnetics

In ECE 331, the boundary conditions for Maxwell's equations were determined in the electrostatic and magnetostatic cases by considering the loops and cylinder shown in the following diagrams.

The differences in the 2 additional terms $\frac{\partial \mathbf{B}}{\partial t}$ and $\frac{\partial \mathbf{D}}{\partial t}$ in Maxwell's equations when the electric and magnetic fields change with time are 0 in the limits as the areas of the loops go to 0.

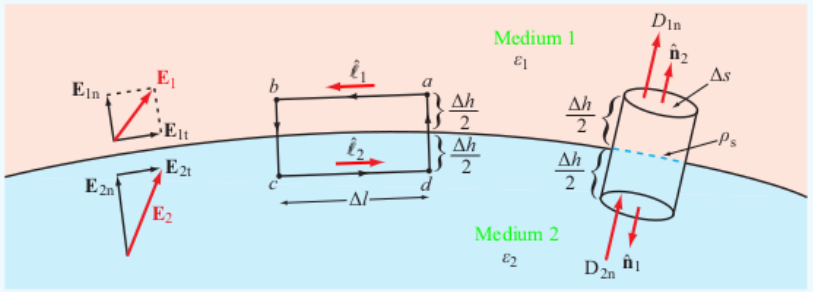


Figure 4-18 Interface between two dielectric media.

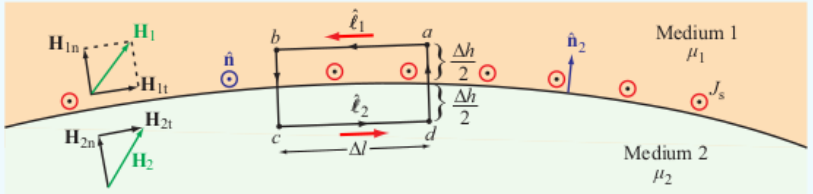


Figure 5-24 Boundary between medium 1 with μ_1 and medium 2 with μ_2.

This means that **the boundary conditions for the electromagnetic fields are the same as in the electrostatic and magnetostatic cases**.

Boundary Conditions for Electromagnetics (Continued)

The boundary conditions are summarized in the following table.

Table 6-2 Boundary conditions for the electric and magnetic fields.

Field Components	General Form	Medium 1 Dielectric	Medium 2 Dielectric	Medium 1 Dielectric	Medium 2 Conductor
Tangential E	$\hat{\mathbf{n}}_2 \times (\mathbf{E}_1 - \mathbf{E}_2) = 0$	$E_{1t} = E_{2t}$		$E_{1t} = E_{2t} = 0$	
Normal D	$\hat{\mathbf{n}}_2 \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s$	$D_{1n} - D_{2n} = \rho_s$		$D_{1n} = \rho_s$	$D_{2n} = 0$
Tangential H	$\hat{\mathbf{n}}_2 \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s$	$H_{1t} = H_{2t}$		$H_{1t} = J_s$	$H_{2t} = 0$
Normal B	$\hat{\mathbf{n}}_2 \cdot (\mathbf{B}_1 - \mathbf{B}_2) = 0$	$B_{1n} = B_{2n}$		$B_{1n} = B_{2n} = 0$	

Notes: (1) ρ_s is the surface charge density at the boundary; (2) $\mathbf{J}_s$ is the surface current density at the boundary; (3) normal components of all fields are along $\hat{\mathbf{n}}_2$, the outward unit vector of medium 2; (4) $E_{1t} = E_{2t}$ implies that the tangential components are equal in magnitude and parallel in direction; (5) direction of $\mathbf{J}_s$ is orthogonal to $(\mathbf{H}_1 - \mathbf{H}_2)$.

Charge-Current Continuity and Kirchhoff's Law

Consider an arbitrary volume $\mathcal{V}$ enclosed by a surface S and let Q be the charge enclosed in $\mathcal{V}$. Then

$$Q = \int_{\mathcal{V}} \rho_v \, d\mathcal{V}. \tag{16}$$

Since **charge is conserved**, i.e. never created or destroyed, ρ_v only changes when charge flows into or out of $\mathcal{V}$. The flow of charge out of $\mathcal{V}$ is the current

$$I = -\frac{dQ}{dt} = -\frac{d}{dt}\int_{\mathcal{V}} \rho_v \, d\mathcal{V}. \tag{17}$$

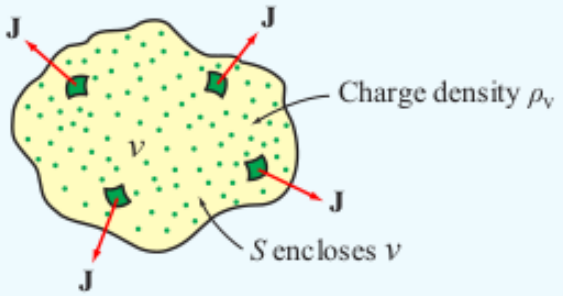


Figure 6-14 The total current flowing out of a volume $\mathcal{V}$ is equal to the flux of the current density $\mathbf{J}$ through the surface S, which in turn is equal to the rate of decrease of the charge enclosed in $\mathcal{V}$.

This current corresponds to a current density $\mathbf{J}$ flowing through the surface S and

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = -\frac{d}{dt}\int_{\mathcal{V}} \rho_v \, d\mathcal{V}. \tag{18}$$

Charge-Current Continuity and Kirchhoff's Law (Continued)

The divergence theorem implies

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{J}\, d\mathcal{V} = \oint_S \mathbf{J} \cdot d\mathbf{s} = -\frac{d}{dt} \int_{\mathcal{V}} \rho_v\, d\mathcal{V}. \tag{19}$$

If $\mathcal{V}$ is fixed, then the time derivative on the right-hand side only applies to ρ_v and

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{J}\, d\mathcal{V} = \int_{\mathcal{V}} -\frac{\partial \rho_v}{\partial t}\, d\mathcal{V}. \tag{20}$$

Finally, this equation must be true for any fixed volume $\mathcal{V}$. Therefore, the integrands must be equal, i.e.

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}. \tag{21}$$

This is the **charge-current continuity equation**.

Charge-Current Continuity and Kirchhoff's Law (Continued)

If the charge flowing into $\mathcal{V}$ matches the charge flowing out of $\mathcal{V}$, then

$$\frac{\partial \rho_v}{\partial t} = 0 \tag{22}$$

and

$$\nabla \cdot \mathbf{J} = 0. \tag{23}$$

These are the conditions in a conductor. Under these conditions,

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = \int_{\mathcal{V}} \nabla \cdot \mathbf{J} \, d\mathcal{V} = 0. \tag{24}$$

The equation

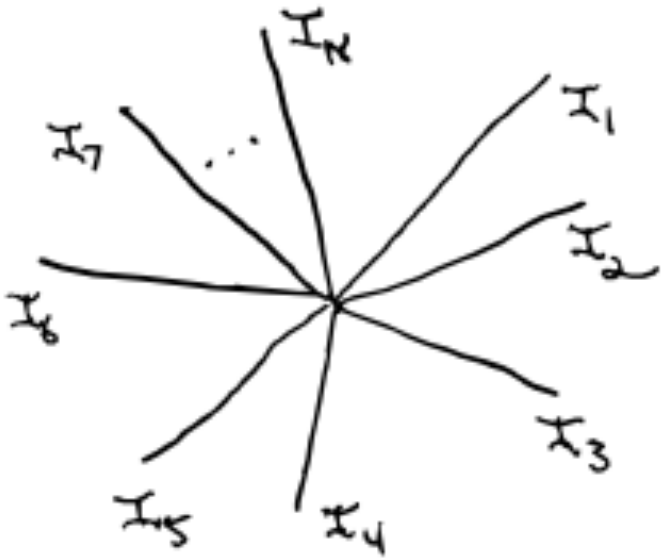
$$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0 \tag{25}$$

is **Kirchhoff's current law**.

Charge-Current Continuity and Kirchhoff's Law (Continued)

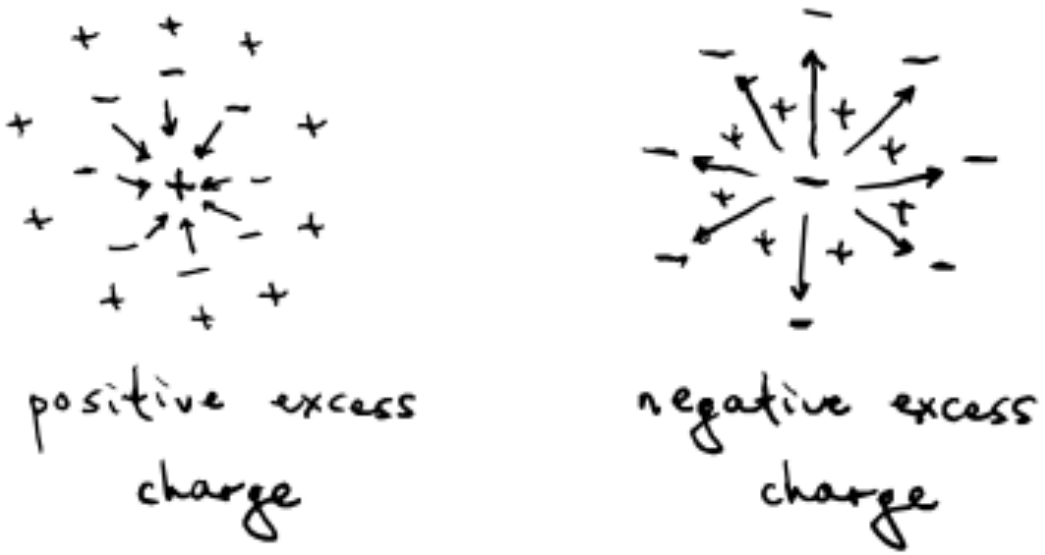
Suppose there are N branches carrying currents $I_1, I_2, \cdots, I_N$ that meet at a junction in a circuit. Consider a volume $\mathcal{V}$ that encloses the junction. For this volume, Kirchhoff's law takes the form

$$\sum_{i=1}^{N} I_i = 0. \tag{26}$$



Free-Charge Dissipation in a Conductor

Conducting materials consist of neutral atoms with relatively loosely-bound electrons. If excess charge q is located within a conducting material, then it emits an electric field that causes the electrons in the material to move toward or away from the charge, until the interior of the material is again neutral. Since the electrons are bound to the atoms in the material, they can't move beyond the material's surface and the end result is that the surface of the material will have a charge q while the interior is neutral.



Free-Charge Dissipation in a Conductor (Continued)

The charges in the material must obey the **charge-current continuity equation**

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}. \tag{27}$$

The electric field $\mathbf{E}$ that the excess charge emits is related to the resulting current by Ohm's law

$$\mathbf{J} = \sigma \mathbf{E} \tag{28}$$

so that

$$\nabla \cdot \mathbf{J} = \sigma \nabla \cdot \mathbf{E} = \sigma \frac{\rho_v}{\epsilon}, \tag{29}$$

where the last expression comes from Maxwell's equation for $\nabla \cdot \mathbf{E}$. Substituting this into the charge-current continuity equation and re-arranging a little,

$$\frac{\partial \rho_v}{\partial t} + \frac{\sigma}{\epsilon} \rho_v = 0. \tag{30}$$

This equation pertains to the interior of the material and not the surface.

Free-Charge Dissipation in a Conductor (Continued)

This 1st-order differential equation can be solved by separation of variables and has the solution

$$\rho_v(t) = \rho_{v0} e^{-(\sigma/\epsilon)t} = \rho_{v0} e^{-t/\tau_r} \quad (\mathrm{C/m^3}), \tag{31}$$

where $\rho_{v0} = \rho_v(0)$ is the charge density at $t = 0$ and

$$\tau_r = \frac{\epsilon}{\sigma} \tag{32}$$

is the **relaxation time constant**.

Material	$\boldsymbol{\epsilon}$	$\boldsymbol{\sigma}$	$\boldsymbol{\tau_r}$
Copper	$\epsilon_0 = 8.854 \times 10^{-12}$ F/m	5.8×10^7 S/m	1.53×10^{-19} s
Mica	$6\epsilon_0 = 5.312 \times 10^{-11}$	10^{-15} S/m	5.31×10^4 s

For mica, $\tau_r \sim 14.8$ hours. This illustrates the typical difference between a conductor (copper) and a good insulator (mica).

Scalar Electric Potential

In the context of **electrostatics**, the **scalar potential** V was defined by

$$\mathbf{E} = -\nabla V. \tag{33}$$

This was appropriate since the curl of the gradient of a scalar function is always 0 and this definition guarantees that

$$\nabla \times \mathbf{E} = \mathbf{0}, \tag{34}$$

which is **Faraday's law** for electrostatics. However, Faraday's law in the dynamic case is

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{35}$$

and $\mathbf{E} = -\nabla V$ cannot satisfy this equation.

Electromagnetic Potentials (Continued)

Vector Magnetic Potential

Since

$$\nabla \cdot \mathbf{B} = 0 \tag{36}$$

in both the magnetostatic and dynamic cases and the vector potential $\mathbf{A}$ was defined by the equation

$$\mathbf{B} = \nabla \times \mathbf{A} \tag{37}$$

to guarantee that $\nabla \cdot \mathbf{B} = 0$ since the divergence of the curl of a vector field is always 0, the vector potential is still valid in the dynamic case and it's possible to use it to define a potential for $\mathbf{E}$ that is valid in the dynamic case too.

Electromagnetic Potentials (Continued)

Potentials for the Electromagnetic Fields in the Dynamic Case

From Faraday's law and the definition of the vector magnetic potential,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = -\frac{\partial}{\partial t}(\nabla \times \mathbf{A}) = -\nabla \times \frac{\partial \mathbf{A}}{\partial t}. \tag{38}$$

So,

$$\mathbf{0} = \nabla \times \mathbf{E} + \nabla \times \frac{\partial \mathbf{A}}{\partial t} = \nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}\right) \tag{39}$$

since the curl is a linear operator. Then, as before, it's possible to define a scalar potential V such that

$$\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} = -\nabla V. \tag{40}$$

This means

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \tag{41}$$

and

$$\mathbf{B} = \nabla \times \mathbf{A}. \tag{42}$$

Retarded Potentials

In the electrostatic case, for a charge distribution in a volume $\mathcal{V}'$, the scalar potential is given by

$$V(\mathbf{R}) = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\rho_v(\mathbf{R_i})}{R'} \, d\mathcal{V}', \qquad (43)$$

where $\mathbf{R}$ is the position of the observation point, $\mathbf{R}_i$ is the position of the volume element $d\mathcal{V}'$, and $R' = |\mathbf{R} - \mathbf{R}_i|$.

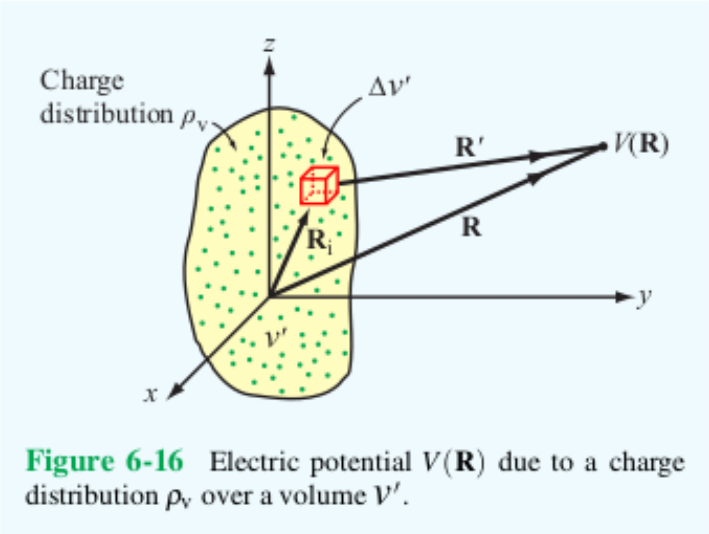


Figure 6-16 Electric potential $V(\mathbf{R})$ due to a charge distribution ρ_v over a volume $\mathcal{V}'$.

If charges move, the effects of a change in the charge at $\mathbf{R}_i$ will not be experienced at $\mathbf{R}$ at the time the change happens. Instead, the effects of the change will propagate outward from $\mathbf{R}_i$ at a finite speed u_p and will reach $\mathbf{R}$ after a delay of R'/u_p.

Retarded Potentials (Continued)

So, when charges move, the scalar potential V at position $\mathbf{R}$ and time t will be

$$V(\mathbf{R}, t) = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\rho_v(\mathbf{R_i}, t - R'/u_p)}{R'} \, d\mathcal{V}'. \tag{44}$$

This is called the **retarded scalar potential** since the effects of a change are *delayed* or *retarded*. The speed of propagation u_p will be the speed of light in the given medium.

Due to similar considerations, when current vary, the vector potential $\mathbf{A}$ at position $\mathbf{R}$ and time t will be

$$\mathbf{A}(\mathbf{R}, t) = \frac{\mu}{4\pi} \int_{\mathcal{V}'} \frac{\mathbf{J}(\mathbf{R_i}, t - R'/u_p)}{R'} \, d\mathcal{V}'. \tag{45}$$

This is the **retarded vector potential**.

Time-Harmonic Potentials

The retarded scalar potential $V(\mathbf{R}, t)$ and the retarded vector potential $\mathbf{A}(\mathbf{R}, t)$ are **linearly dependent** on the charge density $\rho_v(\mathbf{R_i}, t)$ and the current density $\mathbf{J}(\mathbf{R_i}, t)$, respectively:

$$V(\mathbf{R}, t) = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\rho_v(\mathbf{R_i}, t - R'/u_p)}{R'} \, d\mathcal{V}' \tag{46}$$

$$\mathbf{A}(\mathbf{R}, t) = \frac{\mu}{4\pi} \int_{\mathcal{V}'} \frac{\mathbf{J}(\mathbf{R_i}, t - R'/u_p)}{R'} \, d\mathcal{V}' \tag{47}$$

and the electric and magnetic fields $\mathbf{E}$ and $\mathbf{B}$ are **linearly dependent** on the retarded scalar potential and the retarded vector potential, respectively:

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \tag{48}$$

$$\mathbf{B} = \nabla \times \mathbf{A}. \tag{49}$$

This means that the electric and magnetic fields $\mathbf{E}$ and $\mathbf{B}$ are **linearly dependent** on the charge density ρ_v and the current density $\mathbf{J}$.

Time-Harmonic Potentials (Continued)

So, if the charge and current densities are expressed as **superpositions of complex sinusoids**, then it's possible to determine the electric and magnetic fields as superpositions of the effects of individual complex sinusoidal charge and current densities. Therefore, it's worthwhile to determine **E** and **B** when $\rho_v(\mathbf{R_i}, t)$ and $\mathbf{J}(\mathbf{R_i}, t)$ have a complex sinusoidal time dependence. Since the sinusoidal variation is with respect to time, the charge and current densities in this case are called **time-harmonic**. Following this approach to determining **E** and **B** utilizes the **phasor domain** and describes the fields for a single (angular) frequency ω.

A **time-harmonic charge density** has the form

$$\rho_v(\mathbf{R_i}, t) = \rho_v(\mathbf{R_i}) \cos(\omega t + \phi) = \mathrm{Re}\left\{\tilde{\rho}_v(\mathbf{R_i}) e^{j\omega t}\right\}, \tag{50}$$

where $\tilde{\rho}_v(\mathbf{R_i})$ is the phasor of $\rho_v(\mathbf{R_i}, t)$. Recall that sine and cosine differ by a phase shift. So, this is completely general.

Time-Harmonic Potentials (Continued)

Since

$$\begin{aligned}\rho_v(\mathbf{R_i})e^{j\phi}e^{j\omega t} &= \rho_v(\mathbf{R_i})e^{j(\omega t+\phi)} \\ &= \rho_v(\mathbf{R_i})(\cos(\omega t+\phi)+j\sin(\omega t+\phi)),\end{aligned} \tag{51}$$

the phasor includes a term associated with the phase ϕ, i.e.

$$\tilde{\rho}_v(\mathbf{R_i}) = \rho_v(\mathbf{R_i})e^{j\phi}. \tag{52}$$

Let

$$\rho_v^c(\mathbf{R_i}, t) = \tilde{\rho}_v(\mathbf{R_i})e^{j\omega t} \tag{53}$$

be the complex charge density. The retarded scalar potential depends on the retarded charge density and

$$\begin{aligned}\rho_v^c(\mathbf{R_i}, t-R'/u_p) &= \tilde{\rho}_v(\mathbf{R_i})e^{j\omega(t-R'/u_p)} = \tilde{\rho}_v(\mathbf{R_i})e^{-j\omega R'/u_p}e^{j\omega t} \\ &= \tilde{\rho}_v(\mathbf{R_i})e^{-jkR'}e^{j\omega t},\end{aligned} \tag{54}$$

where

$$k = \omega/u_p. \tag{55}$$

Time-Harmonic Potentials (Continued)

Here, k is called the **wavenumber** or **phase constant**.

The complex charge density $\rho_v^c(\mathbf{R_i}, t)$ corresponds to a complex scalar potential

$$\begin{aligned} V^c(\mathbf{R}, t) = \tilde{V}(\mathbf{R})e^{j\omega t} &= \frac{1}{4\pi\epsilon}\int_{\mathcal{V}'} \frac{\rho_v^c(\mathbf{R_i}, t - R'/u_p)}{R'}\, d\mathcal{V}' \\ &= \frac{1}{4\pi\epsilon}\int_{\mathcal{V}'} \frac{\tilde{\rho}_v(\mathbf{R_i})e^{-jkR'}e^{j\omega t}}{R'}\, d\mathcal{V}' \\ &= \frac{1}{4\pi\epsilon}e^{j\omega t}\int_{\mathcal{V}'} \frac{\tilde{\rho}_v(\mathbf{R_i})e^{-jkR'}}{R'}\, d\mathcal{V}'. \end{aligned} \tag{56}$$

Dividing this equation by the common factor of $e^{j\omega t}$ provides a relationship between the phasor of the scalar potential and the phasor of the charge density

$$\tilde{V}(\mathbf{R}) = \frac{1}{4\pi\epsilon}\int_{\mathcal{V}'} \frac{\tilde{\rho}_v(\mathbf{R_i})e^{-jkR'}}{R'}\, d\mathcal{V}'. \tag{57}$$

Time-Harmonic Potentials (Continued)

Due to the linear dependence of the scalar potential on the charge density, the scalar potential corresponding to the original, real-valued, time-harmonic charge density is

$$\begin{aligned} V(\mathbf{R}, t) &= \mathrm{Re}\left\{V^c(\mathbf{R}, t)\right\} \\ &= \mathrm{Re}\left\{\tilde{V}(\mathbf{R})e^{j\omega t}\right\} = \mathrm{Re}\left\{\frac{1}{4\pi\epsilon}e^{j\omega t}\int_{\mathcal{V}'}\frac{\tilde{\rho}_v(\mathbf{R_i})e^{-jkR'}}{R'}\,d\mathcal{V}'\right\}. \end{aligned} \tag{58}$$

A similar discussion shows that the vector potential is given by

$$\mathbf{A}(\mathbf{R}, t) = \mathrm{Re}\left\{\tilde{\mathbf{A}}(\mathbf{R})e^{j\omega t}\right\}, \tag{59}$$

where the phasor $\tilde{\mathbf{A}}(\mathbf{R})$ of $\mathbf{A}(\mathbf{R}, t)$ satisfies

$$\tilde{\mathbf{A}}(\mathbf{R}) = \frac{\mu}{4\pi}\int_{\mathcal{V}'}\frac{\tilde{\mathbf{J}}(\mathbf{R_i})e^{-jkR'}}{R'}\,d\mathcal{V}'. \tag{60}$$

Time-Harmonic Potentials (Continued)

Now that we've found how to determine the phasors of the potentials from the phasors of the charge and current densities, the next step is to determine how to find the phasors of the electric and magnetic fields using the phasors of the potentials.

First, the equations

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{B} = \mu \mathbf{H}, \tag{61}$$

and the fact that the derivatives in the curl are all taken with respect to spatial coordinates imply

$$\tilde{\mathbf{H}} = \frac{1}{\mu} \nabla \times \tilde{\mathbf{A}}. \tag{62}$$

Recall that differentiating by time in the time domain is the same as multiplying by $j\omega$ is the phasor/frequency domain. The differential form of **Faraday's law** in the time domain is

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = -\mu \frac{\partial \mathbf{H}}{\partial t}. \tag{63}$$

Time-Harmonic Potentials (Continued)

So, **Faraday's law in the phasor domain** is

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}} \quad \text{or} \quad \tilde{\mathbf{H}} = -\frac{1}{j\omega\mu}\nabla \times \tilde{\mathbf{E}}. \tag{64}$$

Also, the differential form of **Ampère's law** in the time domain is

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}. \tag{65}$$

In a **nonconducting medium** $\mathbf{J} = 0$, this becomes

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} = \frac{\partial(\epsilon \mathbf{E})}{\partial t} \tag{66}$$

and **Ampère's law in the phasor domain** for a nonconducting medium is

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon\tilde{\mathbf{E}} \quad \text{or} \quad \tilde{\mathbf{E}} = \frac{1}{j\omega\epsilon}\nabla \times \tilde{\mathbf{H}}. \tag{67}$$

Time-Harmonic Potentials (Continued)

The above equations can be used to determine the magnetic field $\mathbf{H}$ if the electric field $\mathbf{E}$ is known or vice versa.

Example: In a nonconducting medium with $\epsilon = 16\epsilon_0$ and $\mu = \mu_0$, if the electric field intensity is

$$\mathbf{E}(z,t) = 10\sin(10^{10}t - kz)\hat{\mathbf{x}} \text{ V/m} \tag{68}$$

find the magnetic field intensity $\mathbf{H}$ and determine the wavenumber k.

The first step is to find the phasor $\tilde{\mathbf{E}}$ of $\mathbf{E}$. The complex-valued field associated with the phasor is

$$\mathbf{E}^c(z,t) = \tilde{\mathbf{E}}(z)e^{j\omega t} = \tilde{\mathbf{E}}(z)\cos(\omega t) + j\tilde{\mathbf{E}}(z)\sin(\omega t). \tag{69}$$

The phasor $\tilde{\mathbf{E}}(z)$ must be chosen so that the real part of $\mathbf{E}^c(z,t)$ is $\mathbf{E}(z,t)$:

$$\tilde{\mathbf{E}}(z)\cos(\omega t) = \mathrm{Re}\left\{\mathbf{E}^c(z,t)\right\} = \mathbf{E}(z,t) = (10 \text{ V/m})\sin(\omega t - kz)\hat{\mathbf{x}}, \tag{70}$$

where

$$\omega = 10^{10} \text{ rad/s}. \tag{71}$$

Time-Harmonic Potentials (Continued)

In this case,

$$\mathbf{E}(z,t) = (10\ \mathrm{V/m})\sin(\omega t - kz)\hat{\mathbf{x}} = (10\ \mathrm{V/m})\cos\left(\omega t - kz - \frac{\pi}{2}\right)\hat{\mathbf{x}}. \quad (72)$$

So,

$$\begin{aligned}\mathbf{E}(z,t) &= \mathrm{Re}\left\{10\cos\left(\omega t - kz - \frac{\pi}{2}\right)\hat{\mathbf{x}} + j10\sin\left(\omega t - kz - \frac{\pi}{2}\right)\hat{\mathbf{x}}\right\}(\mathrm{V/m}) \\ &= \mathrm{Re}\left\{(10\ \mathrm{V/m})\hat{\mathbf{x}}e^{j(\omega t - kz - \pi/2)}\right\} \\ &= \mathrm{Re}\left\{(10\ \mathrm{V/m})\hat{\mathbf{x}}e^{-jkz}e^{-j\pi/2}e^{j\omega t}\right\}. \end{aligned} \quad (73)$$

Here,

$$e^{-j\pi/2} = \cos\left(-\frac{\pi}{2}\right) + j\sin\left(-\frac{\pi}{2}\right) = -j \quad (74)$$

and

$$\mathbf{E}(z,t) = \mathrm{Re}\left\{\left(-j(10\ \mathrm{V/m})e^{-jkz}\hat{\mathbf{x}}\right)e^{j\omega t}\right\}. \quad (75)$$

Time-Harmonic Potentials (Continued)

Therefore,

$$\tilde{\mathbf{E}}(z)e^{j\omega t} = \mathbf{E}^c(z,t) = \left(-j(10\ \mathrm{V/m})e^{-jkz}\hat{\mathbf{x}}\right)e^{j\omega t}, \tag{76}$$

which means

$$\tilde{\mathbf{E}}(z) = -j(10\ \mathrm{V/m})e^{-jkz}\hat{\mathbf{x}}. \tag{77}$$

Using the phasor form of Faraday's law,

$$\begin{aligned}
\tilde{\mathbf{H}} &= -\frac{1}{j\omega\mu}\nabla\times\tilde{\mathbf{E}} = -\frac{1}{j\omega\mu}\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -j(10\ \mathrm{V/m})e^{-jkz} & 0 & 0 \end{vmatrix} \\
&= -\frac{1}{j\omega\mu}\left(-\hat{\mathbf{y}}\frac{\partial}{\partial z}\left(j(10\ \mathrm{V/m})e^{-jkz}\right)\right) = \frac{10\ \mathrm{V/m}}{\omega\mu}\hat{\mathbf{y}}\left(-jke^{-jkz}\right) \\
&= -j\frac{(10\ \mathrm{V/m})k}{\omega\mu}e^{-jkz}\hat{\mathbf{y}}.
\end{aligned} \tag{78}$$

Time-Harmonic Potentials (Continued)

To find k, we'll use the phasor form of Ampère's law to determine an expression for $\tilde{\mathbf{E}}$ and then equate this to the form derived above. Using Ampère's law,

$$\begin{aligned}
\tilde{\mathbf{E}} &= \frac{1}{j\omega\epsilon}\nabla\times\tilde{\mathbf{H}} = \frac{1}{j\omega\epsilon}\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & -j\frac{(10\text{ V/m})k}{\omega\mu}e^{-jkz} & 0 \end{vmatrix} \\
&= \frac{1}{j\omega\epsilon}\left(\hat{\mathbf{x}}\frac{\partial}{\partial z}\left(j\frac{(10\text{ V/m})k}{\omega\mu}e^{-jkz}\right)\right) = \frac{(10\text{ V/m})k}{\omega^2\epsilon\mu}\hat{\mathbf{x}}\left(-jke^{-jkz}\right) \\
&= -j\frac{(10\text{ V/m})k^2}{\omega^2\epsilon\mu}e^{-jkz}\hat{\mathbf{x}}. \qquad (79)
\end{aligned}$$

Equating this to the original expression for $\tilde{\mathbf{E}}$ shows

$$-j\frac{(10\text{ V/m})k^2}{\omega^2\epsilon\mu}e^{-jkz}\hat{\mathbf{x}} = \tilde{\mathbf{E}} = -j(10\text{ V/m})e^{-jkz}\hat{\mathbf{x}}. \qquad (80)$$

Time-Harmonic Potentials (Continued)

Therefore,

$$\frac{k^2}{\omega^2\epsilon\mu} = 1 \quad \text{or} \quad k^2 = \omega^2\epsilon\mu \tag{81}$$

so that

$$\begin{aligned} k &= \omega\sqrt{\epsilon\mu} = \omega\sqrt{16\epsilon_0\mu_0} = 4\omega\sqrt{\epsilon_0\mu_0} = \frac{4\omega}{c} \\ &= \frac{4\times 10^{10}\ \text{rad/s}}{3.00\times 10^{8}\ \text{m/s}} = 133\ \text{rad/m}. \end{aligned} \tag{82}$$

(See the first table in Appendix D on p. 460.)

Now,

$$\begin{aligned} \mathbf{H} &= \text{Re}\left\{\tilde{\mathbf{H}}e^{j\omega t}\right\} = \text{Re}\left\{\left(-j\frac{(10\ \text{V/m})k}{\omega\mu}e^{-jkz}\hat{\mathbf{y}}\right)e^{j\omega t}\right\} \\ &= \frac{(10\ \text{V/m})k}{\omega\mu}\hat{\mathbf{y}}\text{Re}\left\{-je^{j(\omega t-kz)}\right\}. \end{aligned} \tag{83}$$

Time-Harmonic Potentials (Continued)

Continuing the calculation,

$$
\begin{aligned}
\mathbf{H} &= \frac{(10\ \mathrm{V/m})k}{\omega\mu}\hat{\mathbf{y}}\mathrm{Re}\left\{-j(\cos(\omega t - kz) + j\sin(\omega t - kz))\right\} \\
&= \frac{(10\ \mathrm{V/m})k}{\omega\mu}\hat{\mathbf{y}}\mathrm{Re}\left\{\sin(\omega t - kz) - j\cos(\omega t - kz)\right\} \\
&= \frac{(10\ \mathrm{V/m})k}{\omega\mu}\sin(\omega t - kz)\hat{\mathbf{y}} \\
&= \frac{(10\ \mathrm{V/m})(133\ \mathrm{rad/m})}{(10^{10}\ \mathrm{rad/s})\,(4\pi \times 10^{-7}\ \mathrm{H/m})}\sin\left(\left(10^{10}\ \mathrm{rad/s}\right)t - (133\ \mathrm{rad/m})\,z\right)\hat{\mathbf{y}} \\
&= \left(\frac{133}{4\pi} \times 10^{-2}\right)(\mathrm{Vs/Hm})\sin\left(\left(10^{10}\ \mathrm{rad/s}\right)t - (133\ \mathrm{rad/m})\,z\right)\hat{\mathbf{y}} \\
&= 0.106(\mathrm{V}/\Omega\mathrm{m})\sin\left(\left(10^{10}\ \mathrm{rad/s}\right)t - (133\ \mathrm{rad/m})\,z\right)\hat{\mathbf{y}} \\
&= 0.106(\mathrm{A/m})\sin\left(\left(10^{10}\ \mathrm{rad/s}\right)t - (133\ \mathrm{rad/m})\,z\right)\hat{\mathbf{y}}. \qquad (84)
\end{aligned}
$$

Homework

Read Chapter 6.

Homework 1 is posted and is due on Tuesday, April 21st, at 1:30 p.m.